

No. of Functional Features Included in the Solution

Date	July 3, 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Data Preprocessing & Cleaning	Loads creditcard.csv, removes duplicate records, and checks for missing values before modeling.	Notebook (CreditCard.ipynb)	Done	0.5
2	Exploratory Data Analysis (EDA)	Univariate and multivariate analysis including class distribution and a correlation heatmap of features.	Notebook (CreditCard.ipynb)	Done	0.5
3	Feature Engineering	Derives Log_Amount (log-transformed transaction amount) and Hour (from transaction time) as additional model features.	Notebook (CreditCard.ipynb)	Done	0.5
4	Model Training & Comparison	Trains and compares Logistic Regression, Decision Tree, and Random Forest classifiers on the prepared dataset.	Notebook (CreditCard.ipynb)	Done	1
5	Model Persistence	Saves the best-performing Random Forest model as credit_card_model.pkl using joblib for reuse in the web application.	Notebook (CreditCard.ipynb)	Done	0.5
6	Prediction API Endpoint	Flask /predict route (GET/POST) that loads the saved model and returns a prediction, wrapped in try/except error handling.	Backend (app.py)	Done	1
7	Web UI Form	Tailwind CSS glass-morphism form (index.html) plus a card-based stylesheet (style.css) for collecting applicant details.	Frontend (index.html, style.css)	In Progress	0.5
8	End-to-End Integration	Connecting the UI form submission to the Flask /predict endpoint so the trained model drives the displayed result.	Full Stack (Frontend + Backend)	Pending	0.5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	6 (Done)
Core / Must-Have Features	5 (Data Preprocessing, EDA, Model Training, Model Persistence, Prediction API)
Additional / Nice-to-Have Features	3 (Feature Engineering, Web UI Form, End-to-End Integration)
Features Tested & Verified	5

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	1	index.html form, style.css styling
2	Backend / Logic	2	Flask routes (home, predict), model loading via joblib
3	Database / Storage	0	None — dataset read directly from creditcard.csv; no persistent DB
4	API / Integration	1	/predict endpoint (GET/POST) for model inference
5	Security / Authentication	0	Not implemented in current scope